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Radiation and Viscous Dissipation Effects on Laminar Boundary Layer Flow of Nanofluid over a Vertical Plate with a Convective Surface Boundary Condition

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Abstract – This paper focuses on the radiation and viscous dissipation effects on laminar boundary layer flow over a vertical plate with a convective surface boundary condition using different types of nanoparticles. The general governing partial differential equations are transformed into a set of two non-linear ordinary differential equations using the Runge-Kutta method. Two different types of nanoparticles, namely copper water nanofluid and alumina water nanofluid are considered. It is found that the heat transfer coefficient of nanofluids decreases with the increase of Prandtl number or Eckert number, whereas it increases with the increase of radiation parameter.

Keywords - laminar boundary layer, nanofluid, radiation, viscous dissipation.

INTRODUCTION

Natural convection heat transfer is an important phenomenon in engineering systems due to its wide applications in electronic cooling, heat exchangers, and thermal systems. Enhancement of heat transfer in such systems is very essential from the industrial and energy saving perspectives. The low thermal conductivity of conventional heat transfer fluids, such as water, is considered a primary limitation in enhancing the performance and the compactness of such thermal systems. An innovative technique for improvement of heat transfer using nano-scale particle dispersed in a base fluid, known as nanofluid (Choi [1]), has been studied extensively in recent years (Daungthongsuk and Wongwises [2]; Trisaksri and Wongwises [3]) mainly for forced convection applications. Go *et al* [4] analysed the reduced vibration of a microfin array. Maiga *et al* [5] studied numerically the forced convection of Al_2O_3 -water and Al_2O_3 -ethylene glycol nanofluids in a straight tube of circular cross-section using the single phase model. They observed that, for laminar flow, the heat transfer coefficient increases with an increase in the volume fraction of the nanoparticles. For turbulent flow, they observed that, the heat transfer coefficient increases deeply for a very short distance from the inlet section. Moreover they found that the Al_2O_3 -ethylene glycol nanofluid is more effective than to the Al_2O_3 -water nanofluid in heat transfer enhancement.

However, the interaction of radiation and viscous dissipation on the laminar boundary layer flow over a vertical plate in a nanofluid has received little attention. Hence, an attempt is made to analyse the radiation and viscous dissipation effects on the laminar boundary layer flow of nanofluid over a vertical plate with convective surface boundary condition. The governing equations are reduced into a set of ordinary differential equations by means of similarity transformations. The resultant equations are solved using Runge-Kutta fourth order method along with a shooting technique. A parametric study is presented to illustrate the influence of the emerging parameters on the physical quantities.

Recently, Sheikhzadeh *et al* [6] conducted a numerical simulation to investigate the problem of free convection of the TiO_2 -water nanofluid in rectangular cavities differentially heated on adjacent walls. The left and the top walls of the cavities were heated and cooled; respectively, while the cavities right and bottom walls were kept insulated. They found that by increase in the volume fraction of the nanoparticles, the mean Nusselt number of the hot wall increases for the shallow cavities; while, the reverse trend occurs for the tall cavities. Witharana *et al*. [7] has analyzed the stability of nanofluids in quiescent and shear flow fields. Jacob *et al* [8] analysed the boundary layer flow past a stretching/shrinking surface beneath an external uniform shear flow with a convective surface boundary condition in a nanofluid. Hamad and Pop [9] discussed the unsteady MHD free convection flow past a vertical permeable flat plate in a rotating frame of reference with constant heat source in a nanofluid. Emmanuel *et al* [10] discussed the investigations on the nanolayer heat transfer in nanoparticles-in-liquid suspensions. Ayali, Devi and Julie Andrews [11] investigated the laminar boundary layer flow of nanofluid over a flat plate. Tahery *et al* [12] discussed the numerical study of heat transfer performance of homogeneous nanofluids under natural convection. Khaleque and Samad [13] discussed the effects of radiation, heat generation and viscous dissipation on MHD free convection flow along a stretching sheet